

Development Test Challenge: Game SDK and Analytics

Context

Famobi works with browser games that integrate with a shared SDK. We also collect gameplay analytics to understand how games are used and where the player experience can be improved.

For this challenge, you will work with a small browser game and create a simple end-to-end analytics solution around it.

The purpose is to understand how you approach an unfamiliar project, organize a full-stack solution, make technical decisions, and communicate your work.

Expected effort

Please time-box the challenge to approximately 6-8 hours.

We do not expect a production-ready platform. If you do not complete everything within the time-box, submit what you have and explain what remains and how you would continue.

What we provide

- The source code of a small playable browser game located at <https://github.com/Famobi-GmbH/neon-snake>
- Access to the Famobi SDK documentation at <https://docs.famobi.com>

You are not expected to create or redesign the game itself.

The challenge

1. Integrate the Famobi SDK

Integrate the SDK into the provided game by following the Famobi documentation.

The integration should represent the important moments of the game correctly, including loading, starting and ending gameplay, score, progress, and pause behavior when applicable.

Test the integration locally and explain how you confirmed that it works.

2. Add gameplay analytics

Add analytics to the game so that relevant gameplay activity can be recorded.

The collected information should make it possible to understand at least:

- when gameplay starts and ends;
- which level was played;
- whether the player completed, failed, or left the gameplay;
- the player's score or progress;
- when the activity happened.

You may decide which events and data model best represent this information.

3. Build a backend

Create a Node.js backend that receives the gameplay analytics and stores them in Firebase.

You are responsible for defining:

- the API;
- the Firebase data structure;
- how incoming data is checked and handled;
- how the stored data is prepared for the frontend.

The solution only needs to run locally. Deployment is not required.

4. Build an analytics dashboard

Create a React frontend that consumes the backend and presents useful gameplay insights.

The dashboard should include a short overview and at least two meaningful graphs. You may decide which metrics and visualizations provide the most useful picture of the game activity.

The dashboard does not need to be visually elaborate. We are more interested in clarity, usability, and the reasoning behind your choices.

Technical requirements

- Use React for the analytics frontend.
- Use Node.js for the backend.
- Use Firebase to store the analytics data.
- Keep credentials and secrets out of the repository and browser code.
- The complete solution must be runnable locally.

Beyond these requirements, you may choose the libraries, project structure, API design, data model, and testing approach you consider appropriate.

Deliverables

Submit a Git repository containing the game integration, backend, dashboard, and a README. The README should explain:

- how to install and run the project;
- how to test the complete flow locally;
- the main technical decisions you made;
- any assumptions or known limitations;
- what you would improve with more time.

The solution must use the Firebase Local Emulator Suite and must run without access to a real Firebase project. Include all necessary emulator configuration and startup instructions in the repository.

Use of external tools

Please disclose material use in the README and be prepared to explain and modify the submitted code during the follow-up conversation.

Candidate notice

This exercise is used solely to evaluate your application and will not be used as production or commercial work. Do not include secrets, personal data, or third-party proprietary code.

Your submission will be reviewed by the hiring team. The final assessment will involve human review.

If you require an accessibility accommodation or adjustment, please contact frederiko@famobi.com. Use this email to submit the deliverables as well. Good luck!